

LOCALIZATION IN THE COMMUTANT, STAGE 14:

lattices, leaves and ellipsoids.

Klartag's process as a frozen-gap Dirac evolution, walls as sliding lattice points, the resolution ellipsoid of a spectral truncation, and the modular dilation of the non-commutative strata

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Abstract

We take seriously the suggestion that the leaves of the network's wall foliation play the role of Klartag's lattice. We develop the lattice side, the leaf side and the dictionary between them.

- *Klartag's process is spectral.* An ellipsoid free of the points of L is the same as a flat torus $\mathbb{R}^n/L^*$ whose Laplacian has spectral gap at least $4\pi^2$ in the corresponding metric. The contact points are the eigenfunctions at the gap: the first spectral truncation of the Dirac operator. Klartag's constrained Dyson motion evolves the metric while freezing the gap, and it stops at a Voronoi-perfect metric, one that its first spectral truncation determines. Local optima are eutactic, which is John's decomposition.
- *A leaf is a sliding lattice point (proved).* In the probe's precision coordinates, a wall's tangency constraint is exactly Klartag's contact constraint $\langle y \otimes y, dA \rangle = 0$, where y is the wall's point of tangency.
 - A wall is a continuum of lattice points, and only the tangency point binds.
 - Klartag's polyhedron becomes the intersection of infinitely many half-spaces, a convex matrix-fractional set.
 - The layer map $z = Wx$ is Klartag's normalizing transformation. It sends cells to orthants and the input ball to the state ellipsoid, which is the frame the chain already works in.
- *The dilation can never be frozen.* For bias-free networks it is a free direction of every contact configuration (proved), and biases remove it. This is the lattice-side reason for the critical gain mode of stage 13.
- *Spectral truncation is a resolution ellipsoid (proved, checked).* A kink at Mahalanobis distance r has k -th chaos energy $\text{He}_{k-2}(r)^2 \varphi(r)^2 / k!$.
 - Only 5–7% of that energy lies below the Hermite turning point $k = r^2/4$, and essentially none below $r^2/8$.
 - Beyond the turning point the energy decays like $k^{-5/2}$; at $r = 0$ the tail decays like $K^{-3/2}$, measured slope -1.50 .
 - So truncation at chaos order K is the ellipsoid of Mahalanobis radius $2\sqrt{K}$. Walls outside it are invisible to exponential accuracy, and walls inside leave an algebraic tail.
- *The leaf space and its transverse measures.* The Gaussian restricts to transverse measures on every stratum, with the mean as their codimension-one pairing (stage 11). The non-commutative strata are the bent crossings, whose isotropy is $\text{aff}(1)$.

- $\text{aff}(1)$ is isomorphic to the location–scale algebra of stage 12’s hyperbolic upper tier, with the Euler (dilation) generator matching the scale.
- Its group $\text{Aff}_+(1)$ is not unimodular, so the natural transverse weight is not a trace. Its modular flow is generated by the logarithm of the dilation (Takesaki–Haagerup).
- So the critical gain is the modular Hamiltonian of the network’s non-commutative strata.
- On the commutative part, the trace on K_0 of the tile groupoid takes values in the group generated by the cell masses: the network’s gap labels.
- *The code a local probe reveals.* Run Klartag’s sticky evolution with walls as obstacles; its Itô volume accounting checks to 0.3σ . At a maximal resolved probe, the whitened contact normals form a tight frame (John; checked to 10^{-14}). The conic Steiner formula holds in a network cell (checked); its face-dimension law is the global gate-count law. The rows’ Gaussian statistics play Siegel’s role.

We end with what this changes for the chain: why the dilation must be carried, an adaptive resolution rule, positivity as a sticky constraint, and the predictions. The checks are in `scripts/verify_stage14.py`.

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1 The picture in one page

Klartag (lattice)	the network (leaves)
lattice point $x \in L$	a wall, acting through its sliding tangency point x_* (Prop. 3.1)
Ryshkov polyhedron	curved Ryshkov body $\{A : w_a^\top A^{-1} w_a \leq c_a\}$; a bundle of
$\{A : Ax \cdot x \geq 1\}$	cones $\langle w \otimes w, mm^\top - \rho^2 \Sigma \rangle \geq 0$ over the cells
spectral gap of the flat torus	the chaos truncation of the probe's Ornstein–Uhlenbeck
	operator; resolution radius $2\sqrt{K}$ (Cor. 4.2)
contacts = gap eigenfunctions	walls at the resolution radius
frozen-gap Dyson motion of	the sticky localization (Itô accounting checked)
the metric	
perfect form; Voronoi eutaxy	a perfect probe; John's decomposition, so contact normals
	form a tight frame (checked)
termination	never, for bias-free walls: the dilation is always free (Prop. 3.2)
normalizing map T	the layer map $z = Wx$: cells to orthants, the ball to the state
	ellipsoid (the chain's frame)
Siegel's mean value	stage 12's field law; theta series $n(1 + st)^{-1/2}$; the wall density
	at $s = 1$
Rogers' second moment	correlations through the shared state; the leading one is the
	dilation (Mattis) mode
torus $\mathbb{R}^n/L$, Haar	leaf space of the wall foliation; Gaussian transverse measures
	on every stratum
K_0 and its trace	gap labels $\sum_C \mathbb{Z}\gamma(C)$ (cell masses)
space of lattices with scaling	non-commutative strata $C^*(\text{Aff}_+(1))$ with the dilation as
(Bost–Connes)	modular flow (Thm 5.4)

The three new statements.

- **Leaves are sliding lattice points.** A wall imposes Klartag's constraint at every one of its points, and only the tangency point binds. The resolved set is the intersection of the Ryshkov polyhedra of all leaf points.
- **Spectral truncation is a resolution ellipsoid.** The kink at Mahalanobis distance r puts only 5–7% of its chaos energy below the turning point $r^2/4$ (essentially none below $r^2/8$), then decays as $k^{-5/2}$. Order K resolves exactly the walls inside radius $2\sqrt{K}$, with tail $\varphi(r)K^{-3/2}/3\pi$.
- **The dilation is modular.** At bent crossings the isotropy $\text{aff}(1)$ is the upper tier's location–scale algebra. Its group is not unimodular, and the modular flow of its Plancherel weight is the dilation. The critical gain is the modular Hamiltonian of the non-commutative strata, the free direction of every contact configuration, and the conserved charge of the layer map.

For the chain.

1. Carry the dilation (it cannot be pinned).
2. Resolution follows $r^2/4$, so effort follows the wall density.

3. Positivity on the truncated operator system is a truth-free sticky constraint.
4. The cell-adapted probe and its frame code are a local model, not yet an estimator.

2 Klartag's process is a spectral process

2.1 Lattice-free ellipsoids are flat tori with a spectral gap

Let $L \subset \mathbb{R}^n$ be a lattice, A positive definite, and $E_A = \{x : Ax \cdot x < 1\}$. Consider the flat torus $T_A = \mathbb{R}^n/L^*$ with the metric whose inverse is A ($\langle u, v \rangle = u^\top A^{-1}v$). Its Laplace eigenfunctions are the characters $e_\xi(x) = e^{2\pi i \langle \xi, x \rangle}$, $\xi \in L$, with eigenvalues $4\pi^2 A\xi \cdot \xi$, and its volume is $\text{covol}(L)^{-1} \det A^{-1/2}$.

Proposition 2.1 (known; restated). *The following hold.*

- $E_A \cap L = \{0\}$ if and only if $\lambda_1(T_A) \geq 4\pi^2$, and $\text{vol}(E_A) = \kappa_n \text{covol}(L) \text{vol}(T_A)$. So Klartag's theorem says that some flat torus has spectral gap $4\pi^2$ and volume $\gtrsim n^2/(\kappa_n \text{covol} L)$: the packing problem is the spectral-gap problem for flat tori.
- The contacts $\partial E_A \cap L$ are exactly the frequencies of the eigenfunctions at the gap. The first spectral projection $P = \mathbf{1}_{[0, 4\pi^2]}(\Delta_A)$ spans the constants and those characters. Connes–van Suijlekom's truncated operator system $PC(T_A)P$ is spanned by the characters $e_{\xi-\eta}$ with ξ, η contacts: the difference set of the contact code.

2.2 The process as a frozen-gap metric evolution

Klartag's $dA_t = \pi_t(dW_t)$ is a Dyson Brownian motion of the metric, projected so that $d\langle A\xi, \xi \rangle = \langle \xi \otimes \xi, dA \rangle = 0$ for every contact. In spectral words, the metric diffuses while every eigenvalue that reaches the gap stays frozen there.

- **Volume.** By Itô, $d \log \det A_t = \langle A_t^{-1}, \pi_t dW_t \rangle - \frac{1}{2} \sum_{i,j} |\pi_t(u_i \otimes_s u_j)|^2 / (\lambda_i \lambda_j) dt$. The torus volume therefore grows in expectation at a rate set by the number of *unfrozen* metric directions, $\dim F_t = \frac{n(n+1)}{2} - \text{rank}\{\xi \otimes \xi\}$.
- **Termination.** The process stops when $F_t = \{0\}$, i.e. when the contact outer products span Sym_n . This is Voronoi's *perfect* form. In spectral terms, the first spectral truncation determines the metric: *rigidity at finite resolution*.
- **Optimality.** At a local maximum of volume, the KKT conditions give $A^{-1} = \sum \lambda_\xi \xi \otimes \xi$ with $\lambda_\xi > 0$. This is Voronoi's *eutaxy*, which in John position reads $\sum \lambda_\xi u_\xi \otimes u_\xi = \text{Id}$: a tight frame. Voronoi's theorem says the locally densest lattices are exactly the perfect and eutactic ones.
- **Compactness.** By Mahler's criterion, tori with gap at least $4\pi^2$ and bounded volume form a compact set. The process lives in the compact thick part of the space of lattices, and the contacts are its boundary.

Check. A_2 has 6 minimal vectors whose outer products have rank 3 = $\dim \text{Sym}_2$ (perfect), with $\sum x \otimes x = 3 \text{Id}$ (eutactic). $\mathbb{Z}^2$ has 4 minimal vectors of rank 2, so it is not perfect.

Why this matters for us. Stated spectrally, Klartag’s mechanism has four parts: a metric (Dirac) evolution; a resolution threshold (the gap); objects at the threshold that freeze (contacts); and a volume accounting by unfrozen directions. Our setting has all four. The metric is the localization probe’s covariance, the threshold is the chaos truncation (Section 4), and the objects at the threshold are walls (Section 3).

3 Leaves as sliding lattice points

3.1 A wall is a continuum of lattice points

A *probe* is a Gaussian localization state, written as an ellipsoid $E(m, A) = \{x : A(x-m) \cdot (x-m) < 1\}$. Here A is the precision at the chosen Mahalanobis radius. The probe is *resolved* for a wall $H = \{w \cdot x = b\}$ if $E \cap H = \emptyset$, and in *contact* if H is tangent to ∂E .

Proposition 3.1 (the leaf as a sliding lattice point; proved). *1. $E \cap H = \emptyset$ if and only if $\langle (x - m) \otimes (x - m), A \rangle \geq 1$ for every $x \in H$. The wall therefore imposes Klartag’s lattice-point constraint once for each of its points. The binding point is the tangency point $x_* = m + A^{-1}w \frac{b-w \cdot m}{w^\top A^{-1}w}$, and the inequality collapses to $w^\top A^{-1}w \leq (w \cdot m - b)^2 =: c$.*

2. At a contact, with fixed centre, the infinitesimal constraint is $\langle (x_ - m) \otimes (x_* - m), dA \rangle = 0$. This is Klartag’s contact constraint, with the lattice point replaced by the wall’s tangency point, which slides as A moves.*

3. The resolved set $\mathcal{R}(m) = \{A : w_a^\top A^{-1}w_a \leq c_a \ \forall a\}$ is convex (matrix-fractional). It is the intersection of the Ryshkov polyhedra of all points of all leaves: a curved Ryshkov body.

Proof. The A -distance from m to H is $|w \cdot m - b|/\sqrt{w^\top A^{-1}w}$, attained at x_* . Differentiating, $d(w^\top A^{-1}w) = -\langle A^{-1}ww^\top A^{-1}, dA \rangle$, and $A^{-1}w \propto x_* - m$. Finally, $w^\top A^{-1}w$ is convex in $A \succ 0$. $\square$

This is the precise sense in which leaves play the lattice. A lattice point is a leaf of dimension 0. A wall is a leaf of dimension $n - 1$ that acts through one moving point, its point of tangency. Klartag’s polyhedron, cut by finitely many fixed points near the origin, becomes a convex body cut by a family of sliding points.

3.2 With the centre free: a cone in the space of states

In terms of the covariance Σ at Mahalanobis radius ρ , a probe (m, Σ) is resolved for H iff $\langle w \otimes w, mm^\top - \rho^2 \Sigma \rangle - 2bw \cdot m + b^2 \geq 0$. For the bias-free first layer this is

$$\langle w_a \otimes w_a, M \rangle \geq 0 \quad \forall a, \quad M = mm^\top - \rho^2 \Sigma.$$

This is a polyhedral cone in the variable M , and the rank-one matrices $w_a \otimes w_a$, the rows of W_1 , play the lattice vectors. At depth l the walls are piecewise linear. Inside a cell, w_a is replaced by the gated Jacobian row $\nabla z_{l,a}$ (and b by 0, by homogeneity). The resolved set is therefore a *bundle* of such cones over the cells: the lattice varies along the leaf space of the tile groupoid.

3.3 The layer map is Klartag’s normalizing transformation

Klartag ends by applying T with $T(E) = \text{ball}$, which turns the lattice into a packing. Our network applies the dual move at every layer:

- $z = W_1 x$ sends the walls to the coordinate hyperplanes, the cells to the 2^n orthants, and the input ball to the ellipsoid $N(0, W_1 W_1^\top)$;
- in z_l -coordinates the layer- l walls are again coordinate hyperplanes, and the state is the closure’s $N(\mu_l, C_l)$.

So the chain lives in the frame where the “lattice” is fixed and standard (the coordinate arrangement, whose face lattice is the Boolean lattice of gate patterns) and the ellipsoid moves. The fields $r_a = \mu_a / \sqrt{C_{aa}}$ are the Mahalanobis distances to the walls, and contacts are the neurons with $|r_a| = \rho$.

3.4 The dilation can never be frozen

Proposition 3.2 (proved). *For bias-free walls, the dilation $(\delta m, \delta \Sigma) = (m, 2\Sigma)$ preserves every resolution inequality and every tangency, because each constraint is homogeneous of degree 2 in $(m, \Sigma^{1/2})$. It therefore lies in the free subspace F of every contact configuration. More generally, every affine vector field that preserves the contact walls acts inside F . With biases ($b \neq 0$) the dilation is not free.*

Check: along the dilation, $dg_a - 2g_a = 0$ exactly for homogeneous walls, and is $O(1)$ for affine walls.

Corollary 3.3 (the lattice-side reason for the critical gain). *With homogeneous walls, Klartag’s process never terminates: the probe keeps diffusing along the log-scale, whatever contacts it has collected. No local (contact) information pins the scale. The chain must therefore carry it as a conserved charge, which is stage 13’s dilation package.*

Conjecture 3.4 (falsifiable). *In networks with biases, the gain mode’s error-transport factor is strictly below 1, with a gap that grows with the bias scale. Biases are lattice points that break the dilation.*

3.5 Depth sheds contacts

Klartag’s ellipsoid accumulates contacts. The network’s state sheds them. As depth grows, the fields spread as $N(0, t_l)$ (stage 12), so the walls leave the resolution ellipsoid. The number still inside radius ρ is $n \operatorname{erf}(\rho / \sqrt{2t_l}) \approx n\rho \sqrt{2/\pi t_l}$, the wall-density law. Cooling is Klartag’s process run backwards.

4 Spectral truncation is a resolution ellipsoid

4.1 The kink chaos law

For a neuron whose pre-activation is $z = \mu + s\xi$ (field $r = \mu/s$), expand $\operatorname{relu}(z)/s$ in the Wiener chaos of ξ . The first chaos carries $\Phi(r)^2$. The rest is the kink.

Theorem 4.1 (the kink chaos law; proved, and checked). *The k -th chaos energy of $\text{relu}(r + \xi)$ is, for $k \geq 2$,*

$$E_k(r) = \frac{\text{He}_{k-2}(r)^2 \varphi(r)^2}{k!}, \quad \sum_{k \geq 2} E_k(r) = \text{Var} \text{relu}(r + \xi) - \Phi(r)^2.$$

By the Plancherel–Rotach asymptotics:

- $E_k(r)$ is exponentially small for $k < r^2/4$, where r lies outside the oscillatory region $|r| < 2\sqrt{k}$ of He_{k-2} ;
- beyond the turning point, averaging over the oscillation, $E_k(r) \approx \frac{\varphi(r)}{\pi k^2 \sqrt{4k - r^2}}$;
- hence the tail beyond order K is $\sum_{k > K} E_k(r) \approx \frac{\varphi(r)}{3\pi} K^{-3/2}$ for $K \gg r^2/4$.

Proof. Stein: $\mathbb{E}[\text{relu}(r + \xi) \text{He}_k(\xi)] = \mathbb{E}[\text{relu}^{(k)}(r + \xi)] = \mathbb{E}[\delta^{(k-2)}(r + \xi)] = \pm \text{He}_{k-2}(r) \varphi(r)$. The energy is the square over $k!$. The asymptotics are those of the normalized Hermite functions $\psi_j(x) = \text{He}_j(x) e^{-x^2/4} / ((2\pi)^{1/4} \sqrt{j!})$. Since $E_k = \psi_{k-2}(r)^2 \varphi(r) / (k(k-1))$, and ψ_j^2 is exponentially small outside $|x| < 2\sqrt{j}$ with oscillation-averaged value $1/(\pi \sqrt{4j - x^2})$ inside (WKB; the oscillator $-\partial_x^2 + x^2/4$ at energy $j + \frac{1}{2}$), the asymptotics follow. $\square$

Checked:

- The sum rule holds to 10^{-7} relative.
- The share of the kink energy below the turning point $k = r^2/4$ is 0.073, 0.056, 0.049 at $r = 4, 8, 12$; below $r^2/8$ it is 0.003, 0, 0.
- The cumulative tail exponent beyond $4r^2$ is -1.52 .
- At $r = 0$ the tail is 4.214×10^{-5} , 1.337×10^{-6} , 4.185×10^{-8} at $K = 10^2, 10^3, 10^4$ (slopes -1.50 , -1.51). The predicted $\varphi(0)K^{-3/2}/3\pi$ gives 1.339×10^{-6} at $K = 10^3$.

4.2 Truncation at order K is the ellipsoid of radius $2\sqrt{K}$

The Ornstein–Uhlenbeck operator of the probe is the number operator: the k -th chaos is its eigenspace with eigenvalue k . It is the Dirac-type operator whose spectral truncation at level K is what a chain truncated at chaos order K keeps. Theorem 4.1 is its Weyl correspondence for kinks.

Corollary 4.2 (resolution ellipsoid; proved). *Truncating at chaos order K is, up to exponentially small errors, the probe ellipsoid of Mahalanobis radius $2\sqrt{K}$.*

- A wall outside it ($r^2 > 4K$) leaves essentially nothing in the retained chaos. Its whole nonlinear content is at most $\varphi(r)/r^3$.
- A wall inside it is captured up to an algebraic tail $\propto \varphi(r)K^{-3/2}$.

The chaos truncation is Klartag’s resolution threshold. The walls inside the $2\sqrt{K}$ -ellipsoid are the “lattice points in the ellipsoid”, and those near its boundary are the contacts.

Corollary 4.3 (a truncation-error law; derived, heuristic in its use). *With stage 12's field law ($r \sim N(0, t_l)$) across neurons, the chaos energy neglected at order K in layer l is*

$$\sum_a \sum_{k>K} E_k(r_a) \approx \frac{n K^{-3/2}}{3\pi} \mathbb{E}\varphi(r) = \frac{n K^{-3/2}}{3\pi} \cdot \frac{\sin(\theta_l/2)}{\sqrt{\pi}}.$$

Raising the truncation from K to $K+1$ removes the fraction $1 - (K/(K+1))^{3/2}$ of it: 46% ($2 \rightarrow 3$), 35% ($3 \rightarrow 4$), 28% ($4 \rightarrow 5$). For comparison, v56's chaos grading (roughly a step from second to third chaos in the κ_4 diagonal) took 21% off the raw MSE. These are the same order, but the MSE has other sources.

At $K=3$ the resolution radius is $2\sqrt{3} = 3.46$. The fraction of neurons inside it is $\text{erf}(\sqrt{2K/t_l})$: 0.92 at $l=8$ ($t \approx 4$) and 0.68 at $l=16$ ($t \approx 12$). This matches stage 12's measurement that the marginal memory vanishes for $|r| \geq 3$.

4.3 The chain's state lives on a truncated operator system

Truncating at order K replaces the commutative algebra of the probe by the operator system $\mathcal{E}_K = P_K L^\infty(\gamma_{m,\Sigma}) P_K$ (Connes–van Suijlekom). The chain's carried statistics are a functional on $\mathcal{E}_K$. For that functional to come from a probability law it must be positive, a truncated moment condition. Positivity defines a convex body whose boundary points are functionals with a kernel. These are the analogue of Klartag's contacts: directions in which the chain's pseudo-law has no room left. Section 7 uses this as a truth-free sticky constraint.

5 The leaf space, its transverse measures, and the modular dilation

5.1 Strata, leaves and transverse measures

Stage 11 set up the wall foliation of input space at depth l . Its leaves are the open faces of the activation tiling: cells (codimension 0), walls (1), and crossings (≥ 2). Its groupoid algebra has a finite composition series over the strata, with subquotients $C^*(\text{isotropy}) \otimes \mathcal{K}$:

- $C_0(\mathbb{R}^j)$ at normal crossings, where the isotropy is abelian;
- $C^*(\text{Aff}_+(1))$ at bent crossings, where a deep wall meets a wall that feeds it.

The Gaussian γ induces a transverse measure on every stratum by disintegration. On a stratum S of codimension j it is the Gaussian surface density on S , with the normal space $\mathbb{R}^j$ as local transversal.

Proposition 5.1 (the mean is leaf-space integration; stage 11, restated).

$$\mathbb{E}f = (2\pi)^{-1/2} \sum_{\text{codim } F=1} \kappa_F \alpha_F,$$

the codimension-one transverse measure paired with the kink cocycle κ_F . The cumulant corrections of order k pair the codimension- j transverse measures, $j \leq k-1$, with higher kink cocycles. Only bent chains carry mixed terms (stage 12: Kikuchi level = codimension). The mean, and every correction to the closure, is an integral over the leaf space.

5.2 The tile groupoid and the network's gap labels

Let R_l be the relation “same cell at depth l ”. It is proper, and $C^*(R_l) \cong \bigoplus_C \mathcal{K}(L^2(C))$, so $K_0(C^*(R_l)) \cong \mathbb{Z}^{(\text{cells})}$.

Proposition 5.2 (proved, with the trace normalized by γ). *The trace induced by the Gaussian transverse measure gives each cell C the weight $\gamma(C)$. Its range on K_0 is the gap-labelling group*

$$\text{Gap}_l = \tau_* K_0(C^*(R_l)) = \sum_C \mathbb{Z} \gamma(C) \subset \mathbb{R}.$$

For every self-adjoint element H of $C^*(R_l)$ (a cell-covariant observable, for instance any function of the gate code), the integrated density of states $\tau(\mathbf{1}_{(-\infty, E]}(H))$ at a spectral gap lies in Gap_l . Refinement with depth gives $\text{Gap}_l \subset \text{Gap}_{l+1}$.

This is the commutative half of the leaf space. Its K-theoretic “lattice” is the module of cell masses, the global code of gate patterns weighted by the Gaussian. Bellissard’s gap labelling for quasicrystals has the same shape, with patch frequencies in place of cell masses.

5.3 The non-commutative strata and their modular dilation

Theorem 5.3 (structural; proved at the Lie-algebra level). *Stage 11 found the isotropy algebra at a bent crossing, $\mathfrak{aff}(1) = \langle E, \vartheta \rangle$ with $[E, \vartheta] = \vartheta$, where E is the Euler (dilation) field and ϑ the bending shear. It is isomorphic to the Lie algebra of the location–scale group $\text{Aff}_+(1) = \{x \mapsto \sigma x + \mu\}$, which acts simply transitively on univariate Gaussians. That group is stage 12’s hyperbolic upper tier. The isomorphism sends E to the scale generator D and ϑ to the translation P ($[D, P] = P$).*

Theorem 5.4 (the modular flow is the dilation; known ingredients, applied). *$\text{Aff}_+(1)$ is not unimodular. Its left Haar measure is $d\sigma d\mu/\sigma^2$, its right Haar measure $d\sigma d\mu/\sigma$, and its modular function $\Delta(\sigma, \mu) = \sigma^{-1}$. The Plancherel weight φ on the group von Neumann algebra $L(\text{Aff}_+(1))$ is faithful, normal and semifinite, but not a trace. Its modular group is (Takesaki, Haagerup)*

$$\sigma_s^\varphi(\lambda_{(\sigma, \mu)}) = \Delta(\sigma, \mu)^{is} \lambda_{(\sigma, \mu)} = \sigma^{-is} \lambda_{(\sigma, \mu)}.$$

The modular Hamiltonian $\log \Delta = -\log \sigma$ is the logarithm of the dilation.

Corollary 5.5 (the critical gain is modular). *On the non-commutative strata, the natural transverse weight is not a trace, and its modular flow is the dilation. The mode that stage 13 found critical (transport factor 1, a conserved charge) is the modular Hamiltonian of the network’s non-commutative strata. Three statements are one: the scale is conserved (homogeneity), never frozen by contacts (Proposition 3.2), and generates the modular flow where the foliation is non-commutative.*

Remark 5.6 (what follows, and what does not). • By Connes’ Thom isomorphism, the group C^* -algebra of $\text{Aff}_+(1)$ has $K_0 = \mathbb{Z}$ and $K_1 = 0$. Gap labels on the bent strata cannot come from a trace. They need the dilation-twisted (KMS) weight, that is, the twisted cyclic cohomology of Connes and Moscovici.

- Tomita–Takesaki gives $JL(G)J = R(G)$, so the commutant of the left group algebra is the right one. Stage 11 had forward activations and backward sensitivities as left- and right-harmonic states. Reading them as the left and right sides of the bent-stratum algebra is an analogy we have not made precise.

5.4 An analogy with the space of lattices

Klartag's process lives, in its dual form, on the space of unimodular lattices $X_n = SL_n(\mathbb{Z}) \backslash SL_n(\mathbb{R})$, in its Mahler-compact thick part. The Bost–Connes and Connes–Marcolli systems make a non-commutative space of lattices with the *scaling* as time evolution. Its KMS states encode the arithmetic, with partition function $\zeta(\beta)$. For the network, the same dilation is the modular flow of the non-commutative strata. This suggests one well-defined object, the *radial partition function*

$$Z_l(\beta) = \frac{\mathbb{E}|h_l(x)|^\beta}{\mathbb{E}|x|^\beta} = \int_{S^{n-1}} |h_l(\hat{x})|^\beta d\sigma(\hat{x}),$$

the Mellin transform of the gain. Its logarithm generates the gain's cumulants: $\frac{d^2}{d\beta^2} \log Z|_0 = \text{Var} \log |h_l(\hat{x})|$, the dilation charge, about $V_l/4$ minus the input radius's share. It is the exact object a dilation-consistent closure must transport (stage 13). The arithmetic content of Bost–Connes has no counterpart here, and we do not claim one.

6 The ellipsoid process on the leaves, and the code it reveals

6.1 The sticky localization

Definition 6.1. Fix a centre m and walls $\{w_a \cdot x = b_a\}$. Start from a resolved probe precision $A_0 = a_0 \text{Id}$. Run Klartag's evolution on the precision,

$$dA_t = \sigma \pi_t(dW_t), \quad \pi_t = \text{projection onto } \{B : \langle y_a \otimes y_a, B \rangle = 0 \text{ for every contact } a\},$$

where $y_a = A^{-1}w_a$ is the direction to wall a 's tangency point (Proposition 3.1). A wall becomes a contact when it first touches the probe, and stays one. The process freezes when the $y_a \otimes y_a$ span Sym_n .

By Itô, $\mathbb{E} \log \det A_T = \log \det A_0 - \frac{\sigma^2}{2} \mathbb{E} \int_0^T \delta_t dt$, with $\delta_t = \sum_{i,j} |\pi_t(u_i \otimes_s u_j)|^2 / (\lambda_i \lambda_j)$. Since $\log \det$ is concave, the expected probe volume grows at a rate set by the free directions. *Check* ($n = 3$, 14 affine walls, 200 paths, $T = 2$):

- $\mathbb{E} \log \det A_T = 18.700 \pm 0.050$, against the Itô prediction 18.683 ± 0.010 , from $\log \det A_0 = 19.114$;
- contacts accumulate to 1.9 on average and up to $6 = \dim \text{Sym}_3$, i.e. perfect.

With homogeneous walls and the centre free, Proposition 3.2 keeps the dilation free forever. The probe then never freezes, and its log-scale performs the free walk that is the critical mode.

6.2 The maximal resolved probe and its code

Proposition 6.2 (Klartag's terminal state meets John; proved, and checked). *The locally maximal resolved probe about m is the polar of the Löwner ellipsoid of the dual wall points $\pm w_a / |w_a \cdot m - b_a|$. It touches the contact walls at Mahalanobis distance 1, and its precision satisfies the eutaxy (John) condition*

$$A = \sum_a \lambda_a x_a \otimes x_a, \quad \lambda_a \geq 0, \quad \sum \lambda_a = n, \quad x_a = \frac{w_a}{|w_a \cdot m - b_a|}.$$

This is the polar form of Voronoi's eutaxy: for lattice points the inverse of the ellipsoid matrix is the frame sum, while for walls the precision is. In the whitened frame $u_a = A^{-1/2}x_a$, $\sum_a \lambda_a u_a \otimes u_a = \text{Id}$: the contact normals form a weighted tight frame, a spherical 2-design.

Check ($n = 4, 9$ walls): Khachiyan’s algorithm gives 4 active walls, all at Mahalanobis distance 1.0000, with $\|\sum \lambda_a u_a \otimes u_a - \text{Id}\| = 1.3 \times 10^{-14}$ and $\sum \lambda_a = 4.0000$.

This is the transformation that turns a local process into a code. Klartag’s T maps his terminal ellipsoid to the ball and his lattice to a packing. Our whitening $A^{1/2}$ (the map $x \mapsto A^{1/2}(x - m)$) maps the maximal resolved probe at an input to the unit ball, and the walls around that input to a tight frame of contact normals. The local geometry of linearity around an input, expressed in its own natural metric, is a spherical design: the neighbourhood of every input carries a small frame code.

6.3 Cells as cones: the conic Steiner formula and the global gate-count law

ReLU is the metric projection onto the orthant, $\text{relu}(z) = \Pi_{\mathbb{R}_+^n}(z)$. Moreau’s decomposition $z = \Pi_K z + \Pi_{K^\circ} z$, i.e. $\text{relu}(z) - \text{relu}(-z) = z$, is the cone–polar duality, our analogue of lattice–dual lattice. For any closed convex cone K and $g \sim N(0, \text{Id})$, the conic Steiner formula (McCoy–Tropp) says

$$\mathbb{E} f(|\Pi_K g|^2, |\Pi_{K^\circ} g|^2) = \sum_k v_k(K) \mathbb{E} f(\chi_k^2, \chi_{n-k}^2),$$

where v_k is the law of the dimension of the face that $\Pi_K g$ lands in (the conic intrinsic volumes).

- For the orthant, v_k is binomial: the face is the gate pattern. For orthogonal weights the first layer is exactly this, so v_k is the global gate-count law.
- *Check*, in a layer-1 cell $C = W^{-1}\mathbb{R}_+^n$ with Gaussian W ($n = 10, 2 \times 10^4$ samples, exact projections by NNLS):
 - $\mathbb{E} |\Pi g|^2 = 3.179 \pm 0.020$ against $\sum k v_k = 3.188$;
 - $\mathbb{E} |\Pi g|^4 = 17.89 \pm 0.24$ against 17.93;
 - $\mathbb{E} |\Pi g|^2 |\Pi^\circ g|^2 = 20.12 \pm 0.16$ against 20.33.
- The cell’s statistical dimension is 3.19, against $n/2 = 5$ for orthogonal weights. Gaussian weights make the cell thinner than an orthant.

6.4 Siegel’s principle for the network’s rows

Klartag’s analysis rests on Siegel’s mean value theorem: a random lattice is, on average, Lebesgue measure. Our analogue is stage 12’s self-localization. Through Gaussian rows independent of the state, the fields r_a of a probe are, across rows, $N(0, t)$ -distributed. The “theta series” of the wall code relative to the probe therefore has the Siegel mean

$$\Theta(s) = \sum_a e^{-s r_a^2/2}, \quad \mathbb{E} \Theta(s) = n (1 + st)^{-1/2}.$$

Its value at $s = 1$ is $\sqrt{2\pi}$ times the wall density, $n/\sqrt{1+t}$, which is stage 12’s wall-density law. Rogers’ second-moment formula has an analogue as well. Given the state, the rows are independent (Poisson-like), and the correlations come only through the shared state. The leading one is the collective (dilation) mode, the Mattis component of stage 12, which is once more the charge.

7 What it changes for the chain

7.1 What the dictionary says about the chain

1. The dilation must be carried, because it cannot be pinned.

- In a bias-free network the dilation is a free direction of every contact configuration (Proposition 3.2) and the modular flow of the non-commutative strata (Theorem 5.4).
- So no closure built from local (contact) data fixes it, and errors along it are not damped. That is note XLIV's critical mode, now seen from three sides.
- The remedy is stage 13's: carry the charge (the radial partition function's second cumulant) and write the variance, κ_3 and κ_4 along the dilation as one package.
- *Prediction:* networks with biases show a damped gain mode.

2. Resolution is set by $r^2/4$, so effort should follow the wall density. By Corollary 4.2, a neuron at field r needs chaos order about $r^2/4$ before its kink is seen at all, and contributes at most $\varphi(r)/r^3$.

- The chain's truncation error at order K concentrates on neurons with $r^2 \lesssim 4K$, weighted by $\varphi(r)K^{-3/2}$.
- *Prediction (testable on note XLI's dumps):* the per-neuron benefit of v56's chaos correction is supported on $|r| \lesssim 2\sqrt{3}$, with a $\varphi(r)$ -weighted profile.
- *Prediction:* the next chaos order removes about 35% of the remaining chaos-truncation energy.

3. Positivity is a sticky constraint. The chain's state is a functional on the truncated operator system $\mathcal{E}_K$. If it leaves the positive cone, the pseudo-law has negative mass somewhere near a wall.

- Klartag's rule is to project each layer's update onto the face it touches and continue there.
- This is truth-free and costs $O(n^2)$ per layer for the marginal (one-neuron) moment matrices.
- A first diagnostic: count how often v56's per-neuron moment matrices (orders ≤ 4) fail to be positive semidefinite.

4. The cell-adapted Gaussian. The maximal resolved probe at an input is the natural local Gaussian of its cell, and its frame code records the cell's local wall geometry. It is a candidate local model for future hybrid estimators. Sampling is roughly $1000\times$ too expensive at this budget (stage 13), so this is no algorithm yet.

7.2 Open problems

1. **Klartag's estimate for walls.** Prove that, for Gaussian rows, the sticky localization reaches a resolved probe of log-volume $\geq c \cdot (\text{free dimensions})$ before contacts saturate. Klartag's n^2 -versus- n gain would be the ratio between the best Gaussian probe and the best ball inside a typical cell.
2. **Twisted gap labels.** Compute the dilation-twisted (KMS) pairing of the bent strata, the non-commutative part of the gap-labelling group, and decide whether it equals a correction term of the mean.

3. **Left and right.** Make precise whether the Tomita conjugation $JL(G)J = R(G)$ at bent strata exchanges stage 11's forward and backward harmonic states.
4. **The radial partition function.** Find a recursion for $Z_l(\beta)$ through a layer. The dilation package of stage 13 uses only its second cumulant.
5. **Codes across inputs.** The tight frames of Proposition 6.2 vary with the input. Is their distribution over the sphere a function of the internal clock alone, as stage 12's field law is?

Verification

The checks are run by `scripts/verify_stage14.py`; its output is saved next to this note.

identity	result
1. Perfect and eutactic (Section 2)	A_2 : 6 minimal vectors, rank 3, $\sum x \otimes x = 3\text{Id}$; $\mathbb{Z}^2$: rank 2 (not perfect)
2. Kink chaos law (Thm 4.1)	sum rule to 10^{-7} ; share below $r^2/4$: 0.073, 0.056, 0.049 ($r = 4, 8, 12$); below $r^2/8$: ≤ 0.003 ; tail exponent -1.52 ; at $r = 0$ tail 1.337×10^{-6} at $K = 10^3$ against $\varphi(0)K^{-3/2}/3\pi = 1.339 \times 10^{-6}$, slope -1.50
3. Maximal resolved probe (Prop. 6.2)	$n = 4, 9$ walls: 4 contacts at distance 1.0000; tight frame to 1.3×10^{-14} ; $\sum \lambda = 4.0000$
4. Dilation freedom (Prop. 3.2)	$dg_a - 2g_a = 0$ exactly for homogeneous walls; $O(1)$ for affine walls
5. Conic Steiner formula	layer-1 cell, $n = 10$: 3.179 ± 0.020 vs 3.188 ; 17.89 ± 0.24 vs 17.93 ; 20.12 ± 0.16 vs 20.33 ; statistical dimension 3.19
6. Sticky localization (Section 6)	$\mathbb{E} \log \det A_T = 18.700 \pm 0.050$ vs Itô 18.683 ± 0.010 ; contacts up to $6 = \dim \text{Sym}_3$

References

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- [2] Stage 12 note: *the network as its own two-tier system* ([notes/stage12](#)).
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